



HACKATHON SUBMISSION · PARTICLE × MAGIC × ARBITRUM

Sell anywhere. Settle with certainty.

Walletless checkout for events, creators, cafés, and pop-ups—one familiar sign-in, one available balance, and settlement made authoritative on Arbitrum.

Magic embedded wallet

Particle Universal Accounts

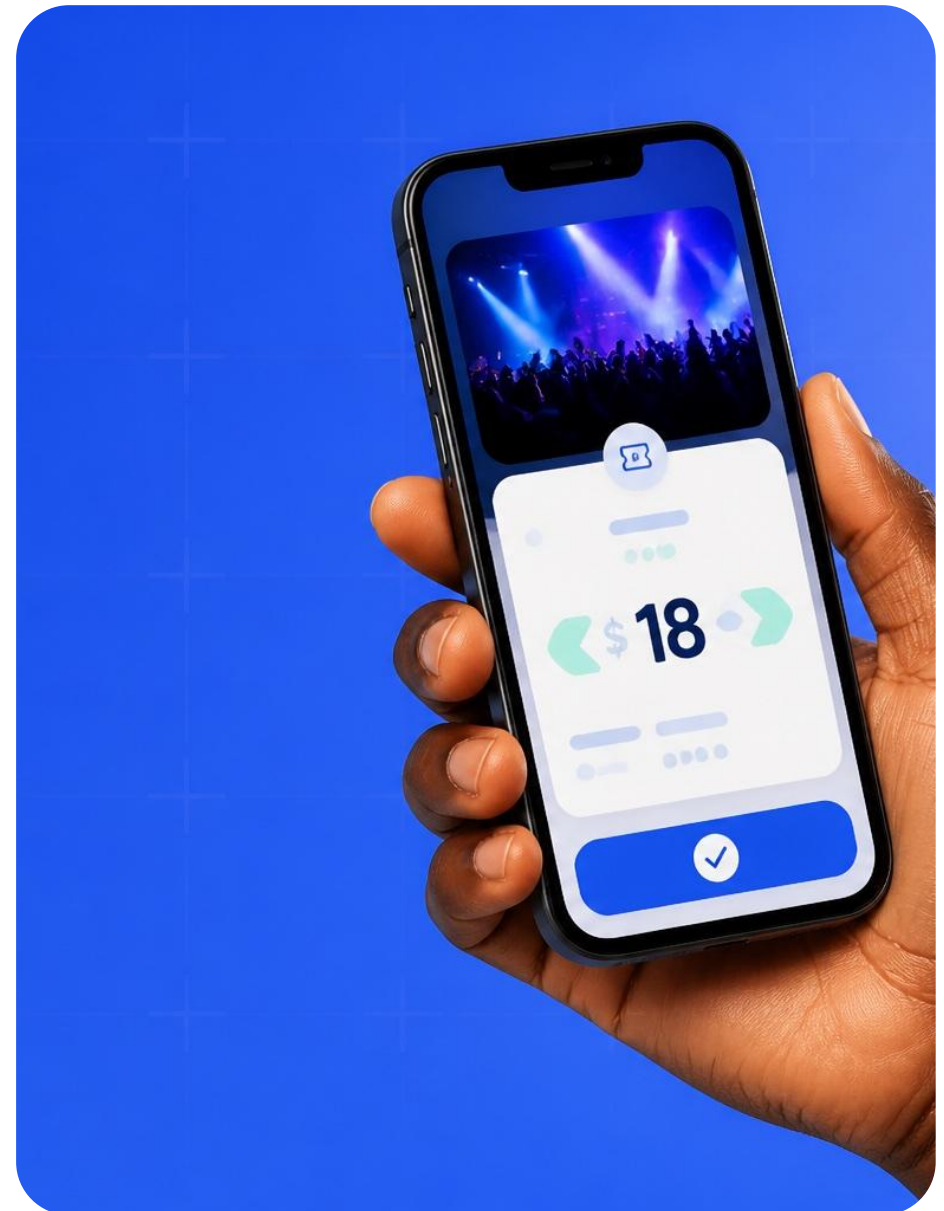
EIP-7702 mode

Arbitrum One settlement

Live application: opentab-opal.vercel.app

Source: github.com/stanleykosi/opentab

Submission source revision:
6828c94edc81ad5b37eeca4bab043ec7300f363c



A checkout customers understand

OpenTab turns a merchant offer into a shareable link or QR. Customers authenticate with Google or email, review a clear total, and approve without installing a browser wallet or manually choosing a payment network.

1

Create

Price,
inventory,
terms,
rewards

2

Share

Public link
or venue QR

3

Sign in

Google or
email with
Magic

4

Review

Item, cost,
maximum
total

5

Confirm

Receipt only
after
settlement

Core truth: a provider response is not payment proof. OpenTab treats a matching, confirmed, canonical Arbitrum OrderPaid event as authoritative.



One offer can be shared online or presented at a physical venue.

Familiar customer journey

No extension, seed phrase, manual bridge, network switch, or destination-gas setup in the ordinary checkout experience.

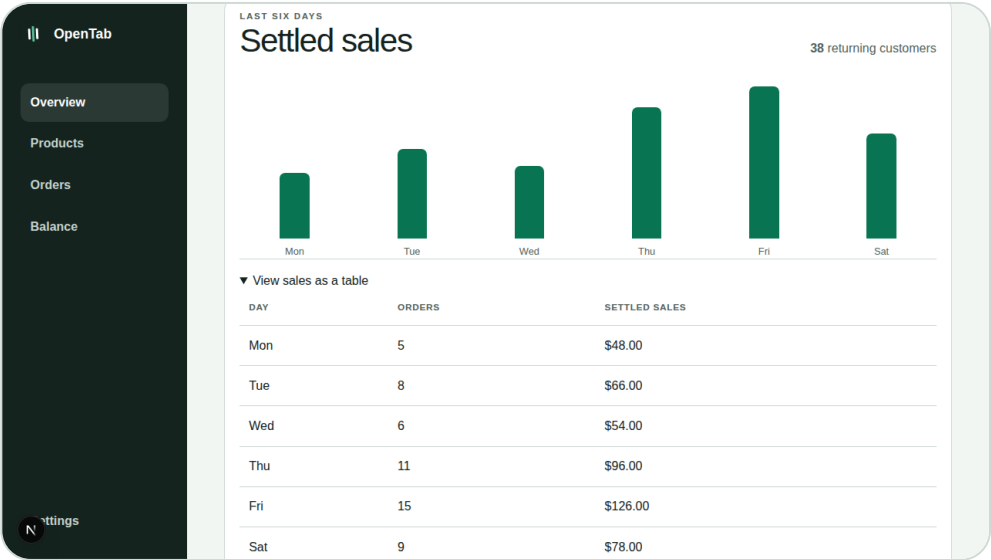
Complete merchant loop

Products, inventory, orders, settlement, refunds, withdrawals, loyalty, analytics, links, and QR sharing.

Proof without clutter

Normal users see calm commerce. A separate public-safe Judge Mode exposes the integration and settlement evidence.

One console from storefront to settlement



Deterministic interface capture. Values are synthetic and shown only to demonstrate product behavior.

Overview

Confirmed revenue, returning customers, recent orders, and settlement freshness at a glance.

Products

Create and manage prices, inventory, availability, purchase limits, event information, checkout links, and QR codes.

Orders and balance

Track pending and confirmed states separately. Only confirmed settlement becomes paid revenue or withdrawable balance.

Settings and operations

Manage storefront information, support details, exports, refunds, withdrawals, and guarded feature controls.

Exact

Integer-based USDC accounting—never JavaScript floating point for token base units.

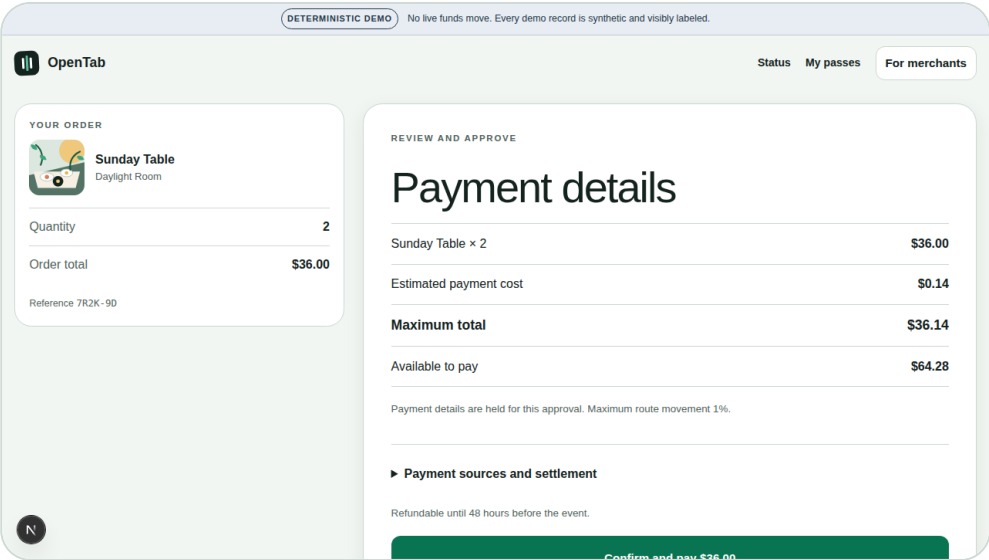
Reorg-aware

The indexer verifies confirmations and canonical block hashes before updating merchant projections.

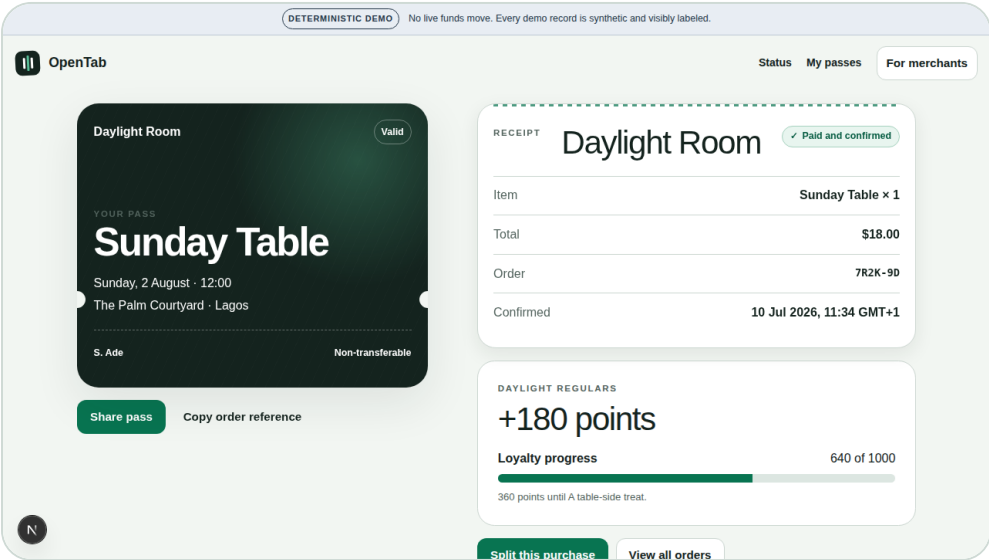
Recoverable

Saved operations and idempotency controls prevent refreshes and repeated clicks from creating duplicate payments.

Familiar access, durable results



Deterministic payment-review capture. The interface shows item cost, estimated payment cost, maximum total, and available balance before approval.



Deterministic receipt/pass capture. No live payment is represented by this screenshot.

Magic authentication

Google or email sign-in provisions the embedded EOA. OpenTab exchanges the DID token for an opaque, server-managed session.

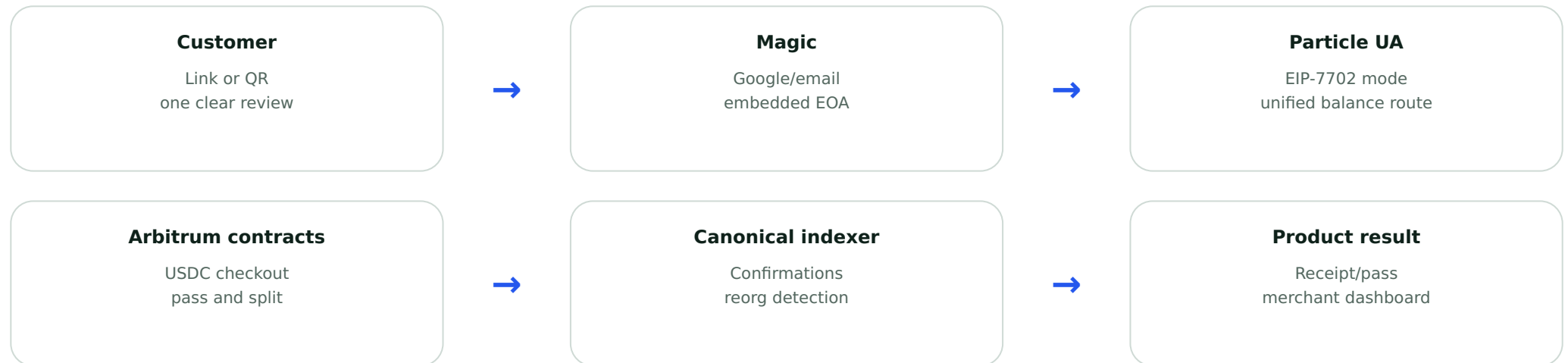
Same-account EIP-7702 design

Particle Universal Accounts is configured with explicit useEIP7702: true; the Magic EOA remains the owner address.

After checkout

Confirmed customers receive a durable receipt/pass, loyalty progress, refund state, and privacy-safe split-reimbursement tools.

Chain abstraction with an authoritative settlement boundary



Arbitrum One deployment

Checkout	0x237E5Da5E0a1F7230E6AE93D737b9cecbcfDee91
Pass	0x56CCBeC6D08f561eCF117964FAB385CBf90A568B
Split	0x7EF7efa8a53530dEa3F077691422AAbEB183049c
Native USDC	0xaf88d065e77c8cC2239327C5EDb3A432268e5831

Deployment facts

- ✓ Arbitrum One chain ID 42161
- ✓ Canonical deployment block 484866936
- ✓ Non-upgradeable Solidity contracts compiled with 0.8.36
- ✓ Checkout, pass, and split deployments independently verified
- ✓ Sourcify creation/runtime status recorded as exact_match
- ✓ Contract payment success depends on exact native USDC accounting and unique order protection

Built as a product, reviewed as payment infrastructure

22/22

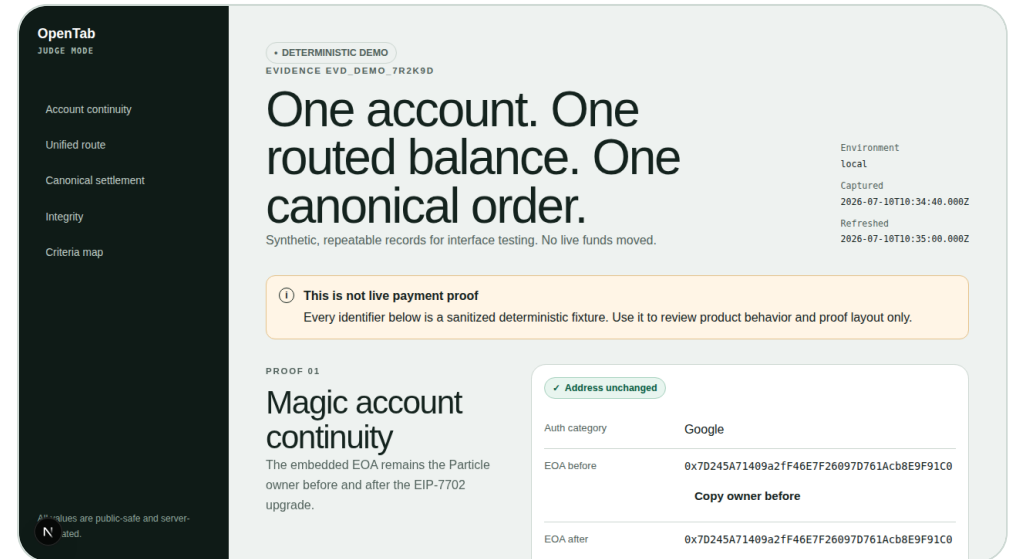
Recorded desktop/mobile Chromium product-journey and quality matrix.

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Ordered PostgreSQL migrations with separate least-privilege runtime identities.

Security and correctness

- ✓ Strict TypeScript and runtime schemas at network and provider boundaries
- ✓ Foundry unit, negative, fuzz, invariant, fork, gas, size, coverage, and Slither evidence
- ✓ Exact integer/string money handling and contract-enforced accounting invariants
- ✓ Opaque sessions, CSRF/origin controls, redacted logs, and guarded feature flags
- ✓ Canonical event reconciliation, restart recovery, idempotency, and reorg handling



Judge Mode interface capture. It is visibly marked deterministic and is not live payment proof.

Try the application

<https://opentab-opal.vercel.app>

Review the source

<https://github.com/stanleykosi/opentab>

Review the contracts

[OpenTabCheckout on Arbiscan](#)